

depends on the separation δ . Therefore, if for each $w \in \mathcal{D}_{W,r}$ the marked points of $\Gamma_{\text{out}}^{*,B(w,2r),B(w,r)}$ are δ -separated, then the conditional probability that the resampling is $\mathfrak{F}_{W,r}$ -successful is positive and uniformly bounded from below.

Lemma 5.4. *Fix $W \subseteq D$ and $r > 0$. For any $M_0 \in \mathbf{N}$, $\delta > 0$, there exists $p > 0$ such that the following is true. Suppose that Γ is a multichordal CLE_κ in $(D; \underline{x}; \beta) \in \mathfrak{D}_{2N}^{\text{ext}}$. Apply the resampling procedure described just above to Γ . Let E_δ be the event that for every $w \in \mathcal{D}_{W,r}$ we have that $\Gamma_{\text{out}}^{*,B(w,2r),B(w,r)}$ is δ -separated. Then for any choice of target pivotals $\mathfrak{F}_{W,r} = \{(\ell_i^1, \ell_i^2, y_i)\}_{i=1,\dots,M^*}$ with $M^* \leq M_0$ we have*

$$\mathbf{P}[G_{W,r}^{\text{res}} \mid \Gamma] \mathbf{1}_{E_\delta} \geq p \mathbf{1}_{E_\delta}$$

where $G_{W,r}^{\text{res}}$ is the event that the resampling is $\mathfrak{F}_{W,r}$ -successful.

Proof. Suppose we are given any target pivotals $\mathfrak{F}_{W,r} = \{(\ell_i^1, \ell_i^2, y_i)\}_{i=1,\dots,M^*}$ with $M^* \leq M_0$. If we perform the resampling operation, then with uniformly positive probability (depending on M_0, r) we sample $\widetilde{M} = M^*$ and the points $w_i \in \mathcal{D}_{W,r/2}$ such that $w_i \in B(y_i, r/2)$ for every $i = 1, \dots, M^*$. In particular, the strands ℓ_i^1, ℓ_i^2 cross the annulus $A(w_i, r, 2r)$. On the event E_δ the strands of $\Gamma_{\text{out}}^{*,B(w_i,2r),B(w_i,r)}$ are δ -separated. Therefore, by Corollary 3.16, the probability that ℓ_i^1, ℓ_i^2 switch their linking pattern after performing the resampling is bounded from below uniformly in δ . Moreover, by continuity (Theorem 1.7) we see that with positive probability bounded from below the new strands $\ell_i^{\text{res},1}, \ell_i^{\text{res},2}$ intersect again inside $B(w_i, r/2)$. (Although we have not phrased the topology (1.2) to keep track of the intersection points between the chords, the proof of Theorem 1.7 shows that continuity also holds when we require the intersection sets to converge.) Repeating this argument M^* times for each target pivotal, we conclude. $\square$

Recall the setup of Section 4.3. Suppose that $D \subseteq \mathbf{C}$ is a simply connected domain and let Γ be a nested CLE_κ in D . Let $u \in (0, 1/8)$ be fixed. Let $j_0 \in \mathbf{Z}$ and $z \in D$ be such that $B(z, 2^{-j_0}) \subseteq D$.

For $j > j_0$, let $U_j = B(z, 2^{-j})$, $V_j = B(z, (1+u)2^{-j})$, and let $\mathcal{G}_j$ be the filtration generated by the partially explored CLE_κ $\Gamma_{\text{out}}^{*,V_j,U_j}$. Let φ_j be the unique conformal map from V_j^{*,U_j} to $\mathbf{D}$ with $\varphi_j(z) = 0$, $\varphi_j'(z) > 0$. Let $(\mathbf{D}; \underline{s}^j; \beta_j)$ be the exterior link pattern decorated marked domain which arises under φ_j . Let $\Gamma_{\underline{s}^j, \beta_j}$ be the image of Γ under φ_j . Conditionally on $\mathcal{G}_j$, we have that $\Gamma_{\underline{s}^j, \beta_j}$ is a multichordal CLE_κ in $(\mathbf{D}; \underline{s}^j; \beta_j)$.

Fix $5/8 < r_1 < r_2 < 7/8$, and let $W = A(0, r_1, r_2)$. We apply the resampling procedure described above to $\Gamma_{\underline{s}^j, \beta_j}$ with W and $r/100$. Let $(\Gamma_{\underline{s}^j, \beta_j})_{W,r/100}^{\text{res}}$ be the multichordal CLE_κ arising from the resampling, and let $\Gamma_{z,j}^{\text{res}}$ be the CLE_κ obtained from mapping back under φ_j^{-1} . Note that any target pivotals $\mathfrak{F}_{W_{z,j},r}$ for Γ induce some target pivotals $\mathfrak{F}_{W,r/100}$ for $\Gamma_{\underline{s}^j, \beta_j}$ via the map φ_j . If $(\Gamma_{\underline{s}^j, \beta_j})_{W,r/100}^{\text{res}}$ is $\mathfrak{F}_{W,r/100}$ -successful, then $\Gamma_{z,j}^{\text{res}}$ is $\mathfrak{F}_{W_{z,j},r}$ -successful, according to Definition 5.2.

Recalling Lemma 5.4, we now see that Proposition 5.3 is a consequence of the independence across scales Proposition 4.7.

Lemma 5.5. *Fix $\delta_0 > 0$, $r > 0$, and $W = A(0, r_1, r_2)$. Suppose that $(\mathbf{D}; \underline{x}; \beta) \in \mathfrak{D}_{2N}^{\text{ext}}$ is δ_0 -separated as in Definition 4.3 and let Γ be a multichordal CLE_κ in $(\mathbf{D}; \underline{x}; \beta)$. Let $\mathcal{D}_{W,r}$ be as in (5.1). Let E_δ be the event that for every $w \in \mathcal{D}_{W,r}$ we have that $\Gamma_{\text{out}}^{*,B(w,2r),B(w,r)}$ is δ -separated. Then*

$$\mathbf{P}_{(\mathbf{D}; \underline{x}; \beta)}^{\text{CLE}_\kappa}[E_\delta] \rightarrow 1 \quad \text{as } \delta \rightarrow 0$$

at a rate that depends only on δ_0 and r .